

Scalability & Future Plan

Date	26/09/2026
Team ID	SWTID- 2026-6463
Project Name	AI FAQ Assistant API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Keyword-based search only, no vector semantic search	Less accurate results for paraphrased questions	Medium
2	Depends on the external Gemini API	AI generation slows or fails on rate limits or downtime	High
3	Single server instance with no caching or rate limiting	Slower responses under heavy concurrent load	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Node.js server instance	Multiple instances with load balancer, PM2 clustering and Docker
2	Data Storage	Single MongoDB instance	MongoDB Atlas with replica sets and sharding
3	Performance	Direct database query on every request	Redis caching, MongoDB indexes and pagination
4	Security	JWT, bcrypt and role-based checks	Rate limiting, refresh tokens, Helmet and input sanitization

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Vector-based semantic search using embeddings	2-3 Months	More accurate and relevant search results
Phase 3	Admin dashboard with FAQ and query analytics	4-6 Months	Better visibility into user question trends
Phase 4	Multi-language support and chatbot integration	6-9 Months	Wider reach and instant customer support